SPLITTING A UNIVERSALLY EXACT COMPLEX INTO UNIVERSALLY EXACT SHORT EXACT SEQUENCES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Right exactness of the tensor product, and the definition of universal exactness by tensoring with every module. Standard module theory, used at proof lines 2772-2776. Supplies the two exact sequences whose combination gives the result.

Lemma 0.1. *Let A be a ring.*

(1) *Let $L \to M \to N$ be a universally exact sequence of A-modules. Let $K = \mathrm{Im}(M \to N)$. Then $K \to N$ is universally injective.*
(2) *Any universally exact complex can be split into universally exact short exact sequences.*

Proof. Proof of (1). For any A-module T the sequence $L \otimes_A T \to M \otimes_A T \to K \otimes_A T \to 0$ is exact by right exactness of $\otimes$. By assumption the sequence $L \otimes_A T \to M \otimes_A T \to N \otimes_A T$ is exact. Combined this shows that $K \otimes_A T \to N \otimes_A T$ is injective.

Part (2) means the following: Suppose that $M^\bullet$ is a universally exact complex of A-modules. Set $K^i = \mathrm{Ker}(d^i) \subset M^i$. Then the short exact sequences $0 \to K^i \to M^i \to K^{i+1} \to 0$ are universally exact. This follows immediately from part (1). $\square$

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